

Rigol MSO5074 CSV Export Analysis

Analysis of oscilloscope exports from a Rigol MSO5074.

- `proben0.csv` — **Save Screen** export
- `proben1.csv` — **Save Memory** export (both above are the **same CH1 acquisition** saved two ways)
- `RigolDS0.bin` — **binary waveform** export (a **separate, later** acquisition; see §8)

Generated by [analyze_rigol.py](#) plus follow-up checks on time/voltage quantization.

1. Side-by-side summary

The two CSV files below are the **same acquisition**; the binary file is a *separate* capture and is analysed on its own in §8 (shown here only for format comparison).

Property	<code>proben0.csv</code> (Save Screen)	<code>proben1.csv</code> (Save Memory)	<code>RigolDS0.bin</code> (Binary)
Header	<code>Time(s),CH1(V)</code>	<code>CH1(V), t0 = -25s, tInc = 2e-06s,</code>	RG01 magic + binary header
Layout	explicit time + voltage per row	voltage only; time = <code>t0 + i · tInc</code>	header + raw float32 buffer
File size	23.45 MB	381.47 MB	3.81 MB
Samples	1,000,000	25,000,000	1,000,000
Start time	-15 s	-25 s	0 s
End time	+15 s	+25 s	50 s
Duration	30 s	50 s	50 s
Sample interval	~30 µs (true)	2 µs (exact)	50 µs (exact)
Sample rate	~33,333 Sa/s (true)	500,000 Sa/s	20,000 Sa/s
Channels	CH1 only	CH1 only	CH1 only

The three sample rates differ because of **acquisition settings (memory depth ÷ timebase)**, not the **file format**. The binary file simply used a lower memory depth (1 Mpts over 50 s = 20 kSa/s).

2. Sample rate — the important correction

Save Memory = 500,000 Sa/s, stated exactly in the header (`tInc = 2e-06 s`).

Save Screen is NOT 10,000 Sa/s. Its true rate is **~33,333 Sa/s** (1,000,000 points spread over 30 s → 30 μs spacing).

The "10 kSa/s / 100 μs" figure that a naive timestamp-difference produces is an **artifact of timestamp rounding** (see below), not the real acquisition rate.

So Save Memory samples roughly **15× faster** than Save Screen (500k vs 33.3k), and stores **25× more points**.

3. What Excel reveals about the Save Screen file

3.1 Time is rounded to 5 significant figures

- The Time(s) column is stored with only ~5 significant digits (-1.5000E+01, -1.4999E+01, ...), i.e. ~10 μs resolution near ±15 s.
- True spacing is ~30 μs, so **3–4 consecutive samples share the same printed timestamp** (e.g. -15.00000 repeated ~17 rows in Excel).
- Of 1,000,000 rows, only **256,669 timestamps are distinct** (avg ≈ 3.9 rows per timestamp).
- Because the smallest representable step is ~10 μs, the median difference between *distinct* timestamps comes out near 100 μs → the misleading "10 kSa/s". This is a rounding effect, not the sampling interval.
- **This rounding is fully recoverable and loses no data** (see §6.7): the samples are uniformly clocked, so the true axis is regenerated by re-indexing as $t[i] = t_0 + i \cdot 30 \mu s$. The rounded column can simply be discarded.

3.2 Voltage is quantized to the ADC step (8-bit LSB)

- The Save Screen file contains only **82 distinct voltage levels**.
- Every value is an integer multiple of **8.7788 mV** — the oscilloscope's 8-bit ADC least-significant bit on this vertical scale.
- 0.0087788 = 1 LSB
- 0.0175576 = 2 LSB
- 0.0351152 = 4 LSB
- 0.0614516 = 7 LSB
- This coarse vertical resolution also clips the signal peaks compared to the Save Memory data.

4. Signal statistics (CH1)

Both exports describe the same signal — DC offset and RMS agree closely.

Metric	Save Screen	Save Memory
Min	-87.79 mV	-272.1 mV
Max	632.1 mV	640.9 mV
Peak-to-peak	719.9 mV	913 mV

Mean (DC offset)	141.6 mV	141.7 mV
RMS	219.3 mV	219.5 mV
Std deviation	167.4 mV	167.6 mV
Dominant freq (FFT)	300.1 Hz	300.1 Hz

Note: once the Save Screen sample interval is computed correctly (points ÷ duration = 30 µs instead of the rounded 100 µs), its FFT also reports **300.1 Hz**, matching the Save Memory file.

5. Key takeaways

- **Save Memory is the high-fidelity export.** ~15× the sample rate, 25× the points, and a wider captured peak-to-peak — it preserves the raw deep-memory waveform.
- **Save Screen is reduced but not corrupted.** It rounds timestamps (5 sig figs) and snaps voltage to the 8.7788 mV ADC grid. Crucially, the timestamp rounding is **fully recoverable** by re-indexing ($t[i] = t_0 + i \cdot \Delta t$), and the samples are **genuine and contiguous** (not peak-detect) — so after reconstruction the Screen file is spectrally faithful for the dominant signatures (see §6.7). Its real limits are possible aliasing (~1.8 % of energy above its 16.7 kHz Nyquist) and the shared 8-bit vertical resolution.
- **Frequency content.** With the corrected time base, both files agree on a dominant component of ~300 Hz and the same harmonic series (50/300/600/900/1500 Hz). (Using the rounded 100 µs timestamp step would wrongly report ~90 Hz for the screen file — another reason to derive the rate from points ÷ duration.)
- When reporting the Save Screen sampling rate, use **points ÷ duration (~33.3 kSa/s)**, not the timestamp-difference value. The script now does this automatically.
- **For quantitative research, prefer Save Memory** for guaranteed alias-free records and the faintest (bearing) signatures; the reconstructed Save Screen file is adequate for RPM, line frequency, harmonics, looseness, transients and broken-rotor-bar analysis.

6. Suitability for scientific fault-detection research

Goal of the study: analyze motor/drive electrical signals to detect **transient processes, running speed (RPM), electrical faults, phase reversal, loose foundation/mechanical looseness, and bearing defects**. This is a **Motor Current Signature Analysis (MCSA)** problem (supply frequency ≈ 50 Hz).

A measurement is fit for purpose only if it satisfies **three independent** requirements. They must be assessed separately — a high sample rate does not compensate for poor amplitude resolution.

Requirement	Set by	Governs
Bandwidth (Nyquist)	sample rate f_s	highest frequency that can be represented ($< f_s/2$)
Frequency resolution Δf	record length T ($\Delta f = 1/T$)	ability to separate closely-spaced sidebands

Dynamic range	ADC bit depth	ability to see faint fault sidebands below the fundamental
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6.1 Frequency content that must be captured (50 Hz machine)

Phenomenon	Where it appears in the current spectrum	Band
Phase reversal	phase sequence / fundamental	50 Hz
Running speed / RPM	1× rotational + harmonics	~10–500 Hz
Mechanical looseness / loose foundation	sub-harmonics (0.5×) and harmonics of 1×, sidebands of 50 Hz	DC—~1 kHz
Broken rotor bars	sidebands $f_s(1 \pm 2ks)$, only a few Hz from 50 Hz	40–60 Hz, sub-Hz spacing
Air-gap eccentricity / rotor-slot harmonics	$f_s \pm m \cdot f_r$ and principal slot harmonics	up to several kHz
Bearing defects (BPFO/BPFI/BSF/FTF)	sidebands $f_s \pm m \cdot f_{\text{bearing}}$ and harmonics	~100 Hz–2 kHz
Startup / transient processes	broadband, time-localized envelope	DC–few kHz

Conclusion on bandwidth: essentially all electrical fault signatures of a line-fed induction machine live **below ~5 kHz**, occasionally to ~10–20 kHz for high-order slot harmonics.

6.2 Empirical capability of each file

	Save Screen (proben0)	Save Memory (proben1)	Requirement
Sample rate	33.3 kSa/s	500 kSa/s	—
Nyquist (max usable freq)	16.7 kHz	250 kHz	$\geq \sim 20$ kHz ✓ both
Record length	30 s	50 s	≥ 10 –20 s for rotor bars
Frequency resolution Δf	0.033 Hz	0.020 Hz	$< \sim 0.5$ Hz ✓ both
ADC depth (measured)	8-bit, 82 levels, 8.78 mV LSB	8-bit, 97 levels, 8.78 mV LSB	≥ 10 –12 bit desired
Faithful contiguous samples?	No — display-decimated, timestamps rounded	Yes	required for transients/spectra

Key measured fact: Save Memory does **not** give more vertical resolution. Both are 8-bit (the MSO5074 has an 8-bit ADC). The Memory export's advantage is **bandwidth and an honest, contiguous time series**, not dynamic range.

6.3 Is the *sample rate* enough?

- **Save Memory — 500 kSa/s: YES, with large margin.** Nyquist 250 kHz vastly exceeds the < 20 kHz needed for MCSA. It is, if anything, *higher than necessary* for current analysis — that headroom could instead be spent on a longer record (finer Δf) or larger memory depth.
- **Save Screen — 33.3 kSa/s: adequate once the time axis is reconstructed.** Bandwidth (16.7 kHz Nyquist) covers MCSA. The export is *reduced* (rounded timestamps, screen-resolution decimation) but **not corrupted**: the rounding is fully recoverable by re-indexing and the samples are genuine/contiguous (verified in §6.7). The remaining concerns are **possible aliasing** of the ~1.8 % of energy above 16.7 kHz (if no anti-alias filter preceded the decimation) and the shared 8-bit resolution — so for the faintest signatures and guaranteed alias-free spectra, **Save Memory is still preferred**.

6.4 Is the *dynamic range* (8-bit) enough? — the real limitation

- Fault sidebands are weak: broken-rotor-bar sidebands are typically **−40 to −60 dBc**, bearing-defect sidebands **−50 to −70 dBc**, relative to the 50 Hz fundamental.
- An ideal 8-bit ADC gives SQNR ≈ **49.9 dB**. Here only ~97/256 codes are used (≈ **6.7 effective bits**, ≈ 42 dB) — the vertical scale was **not optimized**, wasting > 1 bit.
- **Saving grace — FFT processing gain.** With a long record the per-bin noise floor drops by $10 \cdot \log_{10}(N/2)$:
- Memory ($N = 25 \cdot 10^6$): ≈ **71 dB** of gain → spectral floor ≈ −113 dBc;
- Screen ($N = 10^6$): ≈ **57 dB** → spectral floor ≈ −99 dBc.
- So small sidebands **can** be dug out of an 8-bit record **provided** the quantization noise is randomized by real signal noise (natural dither) and the signal is stationary over the record. This is why long records partly rescue the modest bit depth.
- **Net:** dynamic range is **marginal-but-workable** for rotor-bar and eccentricity signatures on the Memory file; it is **weak for bearing detection**, where current signatures are faintest.

6.5 Verdict per phenomenon (using the Save Memory file)

Phenomenon	Verdict	Limiting factor
Phase reversal	■ Trivially sufficient	none
RPM / running speed	■ Sufficient	none
Startup / transient processes	■ Sufficient (true 2 μs samples)	use Memory, never Screen
Mechanical looseness / loose foundation	■ Adequate	amplitude marginal; long record helps
Broken rotor bars	■ Adequate	needs optimized vertical scale + long T
Eccentricity / slot harmonics	■ Sufficient	none
Bearing defects	■■ Marginal via current alone	8-bit dynamic range; prefer vibration/acoustic channels

6.6 Recommendations for the acquisition (act before the next campaign)

1. **Use Save Memory only** for analysis. Prefer the native binary (`.bin/` `.wfm`) over CSV — it is exact, far smaller than the 381 MB CSV, and carries full metadata.

Optimize the vertical scale so the signal fills $\geq 80\%$ of the screen (full ADC range). Today only $\sim 97/256$ codes are used; rescaling recovers

1 effective bit at zero cost.

- 3. **Enable High-Resolution acquisition mode** (boxcar averaging) to gain effective bits and provide built-in anti-alias filtering.
- 4. **Right-size the sample rate.** 500 kSa/s is overkill for current analysis; $\sim 20\text{--}50$ kSa/s already covers slot harmonics. Spending the saved memory on a **longer record** improves Δf , which matters most for rotor-bar sidebands at low slip.
- 5. **Keep records $\geq 10\text{--}20$ s** (both files already do) for sub-Hz resolution to resolve $f_s(1 \pm 2s)$ sidebands.
- 6. **Detect bearings and looseness primarily from the dedicated vibration and acoustic channels** (■■■■■■■■■■/■■■■■■■■■■); current-based bearing detection is inherently weak and is best used only as corroboration.
- 7. **Mind aliasing:** never analyze the Screen/peak-detect export spectrally; ensure anti-alias filtering (hi-res/decimation mode) on the Memory export.

6.7 Re-examined: can we reconstruct the Save Screen time axis, and is the data then enough?

Yes to reconstruction — exactly. The scope samples on a uniform clock, so the true time of sample i is

$$t[i] = t_0 + i \cdot \Delta t, \text{ with } \Delta t = 30 \text{ s} / (N - 1) = 30.00 \text{ }\mu\text{s} \text{ (} N = 1,000,000 \text{)}.$$

The rounded Time(s) column carries **no extra information** — we simply discard it and regenerate the axis from the sample index. (This is precisely what the Save Memory format does: it stores only $t_0 + t_{Inc}$, no per-sample timestamps.) So the 5-significant-figure rounding is a **non-issue for timing** — it is fully recoverable. The earlier "timestamps are rounded" concern is therefore cosmetic, not a loss of data.

Is the Save Screen record then sufficient? Empirically tested — mostly yes for this signal, with two real caveats. Measurements on `proben0` with the reconstructed axis:

Test	Result	Interpretation
Local-extrema fraction	0.079	Smooth, genuine contiguous samples — not peak-detect min/max zig-zag (which would be > 0.6)
Top spectral lines 0–2 kHz	50, 150, 300, 600, 900, 1500 Hz	Matches Save Memory (50, 300, 600, 900, 1500 Hz) — same harmonic structure
Frequency resolution Δf	$1/30 \text{ s} = 0.033 \text{ Hz}$	Fine enough for rotor-bar sidebands
Signal energy within 16.7 kHz Nyquist	98.2 %	Bulk of the signal is inside the Save Screen band

Caveats — what time-axis reconstruction does NOT fix:

- **Aliasing.** 1.78 % of the signal energy (measured on the Memory file) lies *above* the Save Screen Nyquist of 16.7 kHz. If no anti-alias filter preceded the on-screen decimation, that content folds back into the analysis band as a raised noise floor. Small here, but it limits the faintest (bearing) sidebands.
- **8-bit amplitude quantization** is unchanged — still the binding limit for bearing-fault detection.
- **The rate was incidental.** 33.3 kSa/s arose from 1 Mpts ÷ 30 s; a different timebase or memory depth would yield a different (possibly lower) screen rate.

Verdict: with the time axis reconstructed by re-indexing, the Save Screen file **is adequate for RPM, line frequency, harmonics, mechanical looseness, transients and broken-rotor-bar analysis** of this signal. For faint bearing sidebands and a guaranteed alias-free record, **still prefer Save Memory.**

6.8 Bottom line

- **Sample rate is *not* the bottleneck.** Save Memory's 500 kSa/s is more than enough for every electrical motor-fault signature; Save Screen's 33 kSa/s *rate* also suffices, and — with the time axis reconstructed by re-indexing — its samples are genuine and spectrally faithful for the dominant signatures. Its residual weaknesses are **possible aliasing** (~1.8 % out-of-band energy) and the shared 8-bit resolution, so for the faintest faults Save Memory remains preferable.
- **The true limitation common to both is the 8-bit vertical resolution.** It is adequate (helped by long-record processing gain) for rotor-bar, eccentricity, looseness and transient/RPM work, but **marginal for bearing-fault detection** from current — for which the vibration/acoustic channels should lead.

7. Lab procedure: improving vertical resolution & exporting properly

Send this section to the team. It is written for the **Rigol MSO5074** (8-bit ADC, 8 GSa/s, up to 100 Mpts). The goal is to convert the modest 8-bit front-end into an effective **10–12 bits** and to export data that is exact and analysis-ready. Firmware menu names follow UltraVision II; wording may differ slightly between firmware versions.

7.1 Background — why we only had ~6.7 effective bits

The scope digitizes with an **8-bit ADC (256 codes)**. Resolution is fixed in *codes*, not volts, so the only way to gain real bits is to (a) make the signal fill more of those codes, and (b) average away quantization noise. In our files only ~82–97 of 256 codes were used → ~6.7 effective bits. The steps below recover the wasted bits and add more.

7.2 Part A — Fill the ADC range (free ~1–2 bits)

This is the single most important fix and costs nothing.

To Adjust the Vertical Scale

Vertical scale indicates the voltage value per grid in the vertical axis of the screen. It is often expressed in V/div. While you adjust the vertical scale, the display amplitude of the waveform would enlarge or reduce. The scale information of the channel status label (e.g. as shown in the following figure) at the lower section of the screen would change accordingly.

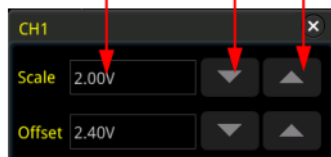


The adjustable range of the vertical scale is related to the currently set probe ratio and input impedance. By default, the probe ratio is 1X and the input impedance is 1 MΩ. The adjustable range of the vertical scale is from 500 μV/div to 10 V/div.

When CH1 is turned on, you can adjust the vertical scale with the following three methods:

- Rotate **Vertical**  **SCALE** that corresponds to CH1 to adjust the vertical scale (clockwise to reduce the scale and counterclockwise to increase).
- Enable the touch screen function and adjust the vertical scale with the pinch & stretch gesture on the touch screen. For details, refer to the "**Pinch & Stretch**" section.
- Enable the touch screen function, and tap the channel status label at the bottom of the screen. The following window is displayed. Tap the   icon at the right side of the **Scale** input field to decrease or increase the scale. You can also tap the **Scale** input field to input a specific value with the displayed numeric keypad.

Value Input Field Decrease Increase



In the vertical control area (Vertical) on the front panel, press **Vertical**  **SCALE** to adjust the vertical scale in "Coarse" (by default) or "Fine" mode. You can also press **1** → **More** → **Fine** to enable or disable fine adjustment.

- Fine adjustment: Rotate **Vertical**  **SCALE** to further adjust the vertical scale within a relatively smaller range to improve vertical resolution. If the amplitude of the input waveform is a little bit greater than the full scale under the current scale and the amplitude would be a little bit lower if the next scale is used, fine adjustment can be used to improve the amplitude of waveform display to view signal details.
- Coarse adjustment (take counterclockwise as an example): Rotate **Vertical**  **SCALE** to set the vertical scale at 1-2-5 step, i.e. 500 μV/div, 1 mV/div, 2 mV/div, 5 mV/div, 10 mV/div...10 V/div.

1. **Connect only the channel(s) you need** and set the correct probe attenuation: CH → Probe (e.g. 1× or 10×) so the displayed volts are true.
2. **Remove DC offset from the screen:** set vertical **Offset/Position** so the signal sits centred, then reduce **Volts/div** (SCALE knob) until the waveform fills $\approx 80\text{--}90\%$ of the screen height *without ever clipping*. - Clipping (flat top/bottom) destroys data — back off one step if you see it. - Include transients/startup surges in the “must not clip” budget; capture a trial run first to learn the true peak.
3. **Use DC coupling** (CH → Coupling → DC) so the full waveform is preserved. Only use AC coupling if a large DC bias forces you to a coarse scale — then AC coupling lets you zoom the AC part for more codes.

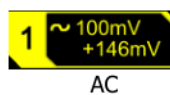
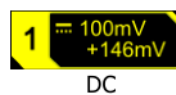


Channel Coupling

The undesired signals can be filtered out by setting the coupling mode. For example, the signal under test is a square waveform with DC offset.

- When the coupling mode is "DC": the DC and AC components of the signal under test can both pass the channel.
- When the coupling mode is "AC": the DC components of the signal under test are blocked.
- When the coupling mode is "GND", the DC and AC components of the signal under test are both blocked.

Press **1** to open the setting menu of CH1. Then, press **Coupling** continuously or rotate the multifunction knob  to select the desired coupling mode (by default, it is DC). The current coupling mode is displayed in the channel status label at the bottom of the screen, as shown in the figure below. You can also enable the touch screen, and then touch the CH1 setting menu to select the desired coupling mode.



Bandwidth Limit

MSO5000 series supports the bandwidth limit function. Setting the bandwidth limit can reduce the noises in the displayed waveforms. For example, the signal under test is a pulse with high frequency oscillation.

- When the bandwidth limit is disabled, the high frequency components of the signal under test can pass the channel.
- When you enable the bandwidth limit and limit it to 20 MHz, 100 MHz, or 200 MHz, the high frequency components found in the signal under test that are greater than 20 MHz, 100 MHz, or 200 MHz are attenuated.

Press **1** to open the setting menu of CH1. Then, press **BW Limit** continuously or rotate the multifunction knob  to select the desired bandwidth limit. Press down the knob to select it. By default, it is OFF. When the bandwidth limit is enabled, the character "**B**" will be displayed in the channel status label at the bottom of the screen,

4. **Verify:** the more screen the trace fills, the more of the 256 codes are used. Filling 80 % vs our 30–38 % recovers more than a full effective bit.

7.2a Choosing the vertical range under variable load / reversal

The captures `1v0.bin` / `500.bin` were taken at **20 % of rated motor power**, where the current is small. The dilemma: if you scale to fill the screen at 20 %, the signal will **clip** during high-current events — startup inrush (typically **5–8× rated current**), overload, or **direction reversal**. Clipping flattens the peaks and **destroys data** (and injects false harmonics into the spectrum). Established practice:

1. **Scale for the worst case, not the present operating point.** The metrology rule: set the vertical range (V/div) to the **maximum expected amplitude** (inrush, reversal, overload), not the momentary level. Do a trial capture of the harshest event first to learn the true peak. Better to "waste" a little resolution at 20 % than to clip during startup.
2. **Split the study by operating point (preferred).** Motor-diagnostics studies almost always use a **series of separate records** — one per steady state (10 %, 20 %, 50 %, 100 % load), each with a scale optimized for *that* point, so every record uses the full ADC range. Capture **transients (startup, reversal) as their own dedicated records** with a wider range. This is the structure already seen in the `■■■■■■■■■■/` folder (1%, 5% ... 100%).
3. **Recover the bits with settings, not range.** Use **High Resolution mode** (§7.3): it yields ~10–12 effective bits by oversampling/averaging, so even a small 20 %-load signal is well resolved on a wide, clip-safe range. This is the best single answer to the small-signal-vs-clipping dilemma. If the current clamp has range settings, pick the one that matches the regime.
4. **Offset shifts, it does not scale.** Centring with vertical offset is fine, but the V/div range must still contain the peaks — offset cannot rescue a range too coarse or too fine.
5. **Move the zero line to the bottom for a unipolar signal.** The 20 %-load current here is almost entirely positive (min $\approx -22 \dots -44$ mV, max $\approx +2.7$ V). On a bipolar, zero-centred scale half the screen (and ~1 effective bit) is wasted. Drop the **0 V reference toward the bottom of the screen** and then **reduce V/div** so the worst-case peak reaches ~90 % at the top → the full ADC range is used upward. Two cautions: (a) offset only moves the window — you must *also* lower V/div to actually gain bits; (b) leave ~1 division below zero so the small negative excursions (min ≈ -44 mV) are not clipped.

Recommendation for this work: for MCSA across loads and reversal, use either (a) a **single fixed range sized for the inrush/reversal peak + High Res** so all records are directly comparable and nothing clips, or (b) a **per-regime range** for best resolution at each point — in which case **record V/div and offset for every file** so the records remain comparable. Avoid clipping at all costs: a clipped record corrupts the spectrum with spurious harmonics that mimic faults.

7.2b DC-motor specifics (this is a *brushed* DC servo machine)

The machine under test is a **brushed permanent-magnet DC servo motor**, type **3PI12.12**, with the following nameplate data:

Parameter	Value
Type	Brushed PM DC servo (commutator + graphite brushes)
Rated power	625 W (0.625 kW)
Rated voltage	110 V DC

Rated current	12.5 A
Rated torque	5.4 N·m
Rated speed	2000 RPM (at 100 % load); ~200 RPM at the ~20 % operating point of these captures
Poles	4 (stator PM excitation)
Commutator segments	~24–36
Drive	External thyristor (SCR) converter, 14XXX series — 4-quadrant, smooth regulation of armature voltage and current ; tachometer/encoder feedback — no built-in driver

The drive is a **thyristor-based converter** (14XXX series), giving smooth regulation of both armature voltage and current. This supports flexible testing of speed/torque strategies, transient regimes and dynamic load variation. It is **not** a PWM (MOSFET/IGBT) drive — which is decisive for interpreting the spectrum below.

Because it is **brushed DC** (not an AC induction machine, and **not BLDC**), the AC-induction "MCSA" framing of §6 does **not** map one-to-one. Account for:

- **The current is unipolar DC + ripple**, not a 50 Hz sinusoid. The armature current carries a DC level (proportional to load/torque) plus ripple. This is exactly why the offset-to-bottom strategy above is so useful here.
- **Keep DC coupling — never AC couple**. The DC level is the signal (it encodes load/torque and reversal). AC coupling would discard it. The earlier "AC coupling to gain codes" tip applies only to AC machines, not here.
- **Two ripple sources to expect in the spectrum:**
- **Thyristor-converter ripple (mains-locked)**. The armature is fed from a **3-phase 6-pulse thyristor converter**, which produces ripple at $6 \times 50 = 300$ Hz plus harmonics (600, 900 Hz ...) on the 50 Hz mains. This is **locked to the mains and independent of motor speed**. At light load / large firing angle the ripple is *deeper* (the conduction notches widen), so it is most prominent exactly in these low-speed captures.
- **Commutation (brush) ripple** at $f_{\text{comm}} = N_{\text{segments}} \times f_{\text{rot}}$, where $f_{\text{rot}} = \text{RPM} / 60$. This is **speed-dependent**: $24...36 \times 33.3 \text{ Hz} \approx 800...1200 \text{ Hz}$ at **2000 RPM (100 %)**, but only $\approx 80...120 \text{ Hz}$ at the **~200 RPM (~20 %)** operating point of these files. This band carries **commutator/brush-wear** information and is a primary brushed-DC diagnostic — make sure the bandwidth/sample rate covers it (16–20 kSa/s easily does).
- **Rotational and its harmonics** at $f_{\text{rot}} = \text{RPM} / 60$ (3.3 Hz at ~200 RPM, 33.3 Hz at 2000 RPM); imbalance/eccentricity/looseness modulate these and the commutation band.
- **Interpreting the measured ~300 Hz**. This is the 6×50 Hz ripple of the **3-phase 6-pulse thyristor converter** — a **supply/drive signature, not a mechanical fault**. The decisive clue: the dominant line is **~300 Hz in every file** even though the captures span different loads/speeds. If it were commutation ripple it would *track speed* (80–120 Hz at 200 RPM, 800–1200 Hz at 2000 RPM); the fact that it stays pinned at 300 Hz proves it is **mains-locked converter ripple**. → Don't mistake it for a defect. To isolate the mechanical/commutation content, **log the actual RPM** (tachometer/encoder) with each record and look in the speed-tracking bands above; optionally notch out 300/600/900 Hz before fault analysis.

- **Reversal on this rig stays unipolar (current never goes negative).** The controller used here **does not drive the armature current below 0 even in reverse** — direction is changed without a negative current excursion. So the “0 at the bottom” unipolar scaling is safe for **all** regimes, including reverse: nothing clips at the bottom. (On a drive that *does* produce negative / regenerative current, you would instead need a bipolar, zero-centred scale — but that is **not** the case here, confirmed by measurement.)
- **Inrush / stall current.** With a 4Q servo drive the start/stall current is limited by the drive's torque (current) limit, but it can still reach several times the 12.5 A rating for short bursts. Scale the shunt/probe so this peak does **not** clip, then trade range back to the working level once you confirm the running current envelope.
- **Faults of interest** on a brushed DC machine: **brush/commutator wear and arcing, bearings, eccentricity, mechanical looseness**, and (for a PM servo) **magnet demagnetization** after over-current. These modulate the DC level, the commutation band and the rotational harmonics.
- **Field vs armature.** This is PM-excited, so there is no separate field winding to monitor — focus on the **armature current**. (If a wound-field variant is ever tested, record field and armature on separate channels.)

7.3 Part B — High-Resolution acquisition mode (adds ~1–4 bits)

Acquire → Mode offers four modes. Choose deliberately:

Acquisition Mode

The acquisition mode is used to control how to generate waveform points from the sample points.

MSO5000 supports the following four acquisition modes: Normal, Average, Peak, and High Resolution. By default, the acquisition mode is Normal.

Press **Acquire** → **Acquisition** on the front panel, and then rotate the multifunction knob  to select the desired acquisition mode. Then, press down the knob to select the mode. You can also press **Acquisition** continuously to select the mode or enable the touch screen to tap the desired mode and select it.

Normal

In this mode, the oscilloscope samples the signal at a specified fixed time interval to rebuild the waveform. For most of the waveforms, using this mode can produce the optimal display effects.

Average

In this mode, the oscilloscope averages the waveforms from multiple samples to reduce the random noise of the input signal and improve the vertical resolution. Greater number of averages can lower the noise and increase the vertical resolution; while at the same time, it will slow the response of the displayed waveform to the waveform changes.

When you select "Average" mode, press **Averages** and rotate the multifunction knob  or use the numeric keypad to set the desired number of averages. You can also use the pop-up numeric keypad to input the average count. When the average count value is not in power-of-2 increments, a prompt message "Truncation average error" will be displayed. At this time, a value that is smaller than the one you input and the closest to power-of-2 increments will be input automatically. For example, if you input 9 with the numeric keypad, the average count will be input 8 automatically.

The number of averages can be set to 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768, or 65536. By default, it is 2.

High Resolution

This mode uses an over-sample technique to average the neighboring points of the sample waveform. This reduces the random noise on the input signal, generates a much smoother waveform on the screen and improves the vertical resolution. This is generally used when the sample rate of the digital converter is greater than the storage rate of the acquisition memory.

Note:

- The "Average" and "High Res" modes use different averaging methods. The former uses "Multi-sample Average" and the latter uses "Single-sample Average".
- In "High Res" mode, the signal bandwidth does not exceed 1/32 of the sampling rate.
- In "High Res" mode, the highest waveform refresh rate mode is not supported.

Sampling Mode

This oscilloscope only supports the real-time sampling mode. In this mode, the oscilloscope produces the waveform display from samples collected during one trigger event. The highest real-time sample rate on the analog channel of MSO5000 series is 8 GSa/s. The current sample rate is displayed under **CH SampleRate**, and you can press **More** → **CH SampleRate** to view it.

Tip

Press **RUN/STOP** on the front panel to stop sampling, and then the oscilloscope will maintain its last captured graph. You can still expand or zoom the waveforms by using the horizontal and vertical control knobs.

Sample Rate

Sampling is the process of converting the analog signal into the digital signal at a specified time interval and then restoring them in sequence. The sample rate is the reciprocal of the time interval.

The sample rate of the analog channel is related to the current channel mode. The maximum real-time sample rate in the single-channel mode of the oscilloscope is 8 GSa/s. The maximum real-time sample rate in the half-channel mode is 4 GSa/s, and the maximum real-time sample rate in the all-channel mode is 2 GSa/s.

Mode	What it does	Use for
------	--------------	---------

Normal	one ADC sample per point	general
Average	averages N repeated acquisitions	stationary, repetitive signals only
High Res	boxcar-averages adjacent samples of a single shot → more bits and a built-in low-pass (anti-alias)	our case — single long records, transients
Peak Detect	keeps min/max — never for analysis	glitch hunting only

Recommended: Acquire → Mode → High Res. - High-Res trades excess sample rate for vertical bits. Since 500 kSa/s is far more than MCSA needs, this is “free” resolution: it can yield **~10–12 bits** while still leaving ≥ 20 kHz of bandwidth. - It also **band-limits before decimation**, which removes the aliasing risk. - Use **Average mode** *only* if the signal is perfectly periodic and the machine state is steady — it does **not** work for transients (startup, load changes).

7.4 Part C — Set rate, memory depth and record length

Tip

Single-channel mode: only one of the four channels (CH1/CH2/CH3/CH4) is enabled.

Half-channel mode: either CH1 or CH2 is enabled; and either CH3 or CH4 is enabled.

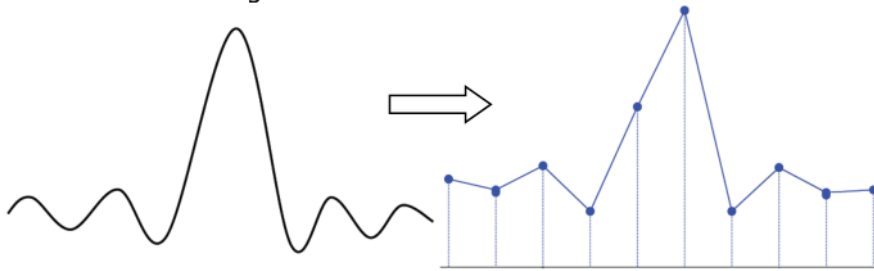
All-channel mode: CH1/CH2 are both enabled or CH3/CH4 are both enabled.

Note that the currently enabled channel should be used as the trigger source at this time.

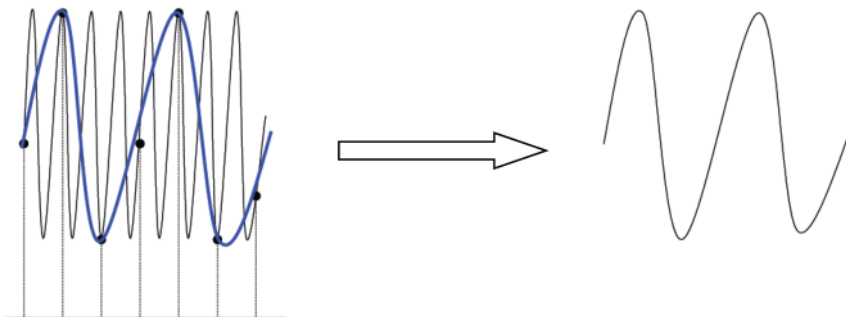
Note: The sample rate is displayed both in the status bar at the upper part of the screen and in the **CH SampleRate** menu. To indirectly modify the sample rate, rotate the **Horizontal**  **SCALE** knob to adjust the horizontal time base or modify the memory depth.

The impact of low sample rate on the waveform:

1. Waveform Distortion: when the sample rate is too low, some waveform details are lost, and the sample waveform displayed is rather different from the actual waveform of the signal.



2. Waveform Aliasing: when the sample rate is twice lower than the actual signal frequency (Nyquist Frequency), the frequency of the waveform rebuilt from the sample data is smaller than the actual signal frequency.



3. Waveform Leakage: when the sample rate is too low, the waveform rebuilt from the sample data does not reflect all the actual signal information.

1. **Acquire** → Mem Depth: choose a **fixed** depth (e.g. 1 Mpts–25 Mpts) rather than “Auto”, so the sample rate is predictable. Displayed $f_s = \text{MemDepth} / \text{timebase span}$.
2. **Right-size the rate.** For 50 Hz MCSA, **20–50 kSa/s** already covers slot harmonics. Prefer spending memory on a **longer record** than on excess rate.
3. **Record length** $\geq 10\text{--}20$ s for sub-Hz frequency resolution ($\Delta f = 1/T$), needed to separate rotor-bar sidebands $f_s(1 \pm 2s)$ at low slip.
4. Set the **horizontal timebase** so total span = desired record length, then confirm the resulting f_s on screen.

7.5 Part D — Export the right way (exact, compact, complete)

Avoid CSV for primary archives — our Save-Memory CSV was **381 MB** and still only 8-bit text. Use the native binary instead.

Storage System

Press **Storage** on the front panel to enter the store and load setting interface. It provides one USB HOST interface on the front panel, which can all be connected to the USB storage device for external storage. The USB storage devices connected are marked as "Removable USB Disk(D)", "Removable USB Disk(E)", "Removable USB Disk(F)", and "Removable USB Disk(G)", respectively.

Storage Type

The file types for storage include image, waveform, and setups. The storage descriptions of each type are as follows.

1. Image

Press **Save Image** → **Format**, and then use the multifunction knob  to select the image type. You can also press **Format** continuously or enable the touch screen to select it. Save the screen image to the internal or external memory in "*.png", "*.bmp", "*.jpg", or "*.tif" format.

After selecting this type:

- Press **File Name** to input the file name to be saved. For filename input, refer to descriptions in "To Create a Folder". Press **Storage** → **More** → **Auto Name** to select "ON", then the filename is generated automatically. If the filename that you input is the same as the existing filename, a prompt message will be displayed, reminding you that the existing file will be overwritten and asking you whether to continue or not.
- Press **Invert** to enable or disable the invert function.
- Press **Color** to select the desired storage color. It can be set to "Gray" or "Color".
- Press **NewFolder** to create a new storage folder. For details, refer to descriptions in "To Create a Folder".
- Press **Window** to display or hide the disk management interface.
- Press **Save** to save the set image file. Press **More** → **Header** to select "ON" or "OFF". If you select "ON", the instrument model, firmware version number, serial number, and the image creation date will be displayed in the header of the image when you save the image file.

Tip

After a USB storage device (FAT32 format, flash memory) is connected, press **Quick** on the front panel to save the file. You can select to save to the internal memory or the external USB storage device. By default, it is saved to the external memory.

2. Wave

Press **Save Wave** to enter the waveform storage setting interface. Save the main setting information (e.g. "On/Off" state of the channel, vertical scale, and horizontal time base) and waveform data of all enabled channels (analog channel and digital channel) to the external memory.

After selecting this type:

- Press **Data Source** to select "Screen" or "Memory".
- Press **Format** to select the format of the waveform storage, such as "*.bin", "*.cvs", and "*.wfm" (when "Memory" is selected under **Data Source**).
- Press **File Name** to input the file name to be saved. For filename input, refer to descriptions in "To Create a Folder". Press **Storage** → **More** → **Auto Name** to select "ON", then the filename is generated automatically. If the filename that you input is the same as the existing filename, a prompt message will be displayed, reminding you that the existing file will be overwritten and asking you whether to continue or not.
- Press **Channel**, and then the available storage channels include CH1-CH4 and D0-D15. Only when you select "Memory" under **Data Source** and the format is "*.bin" or "*.cvs", can you set the storage channel.
- Press **NewFolder** to create a new storage folder. For details, refer to descriptions in "To Create a Folder".
- Press **Save** to save the set waveform file.
- Press **Window** to display or hide the disk management interface.
- Press **More** → **Time information** to enable or disable the display of the time information in the waveform data. If enabled, the time information for the waveform data will be displayed. If disabled, the time information for the waveform data will not be displayed.

3. Setup

Press **Save Setup** to enter the setup storage setting interface. Save the setups of the oscilloscope to the internal or external memory in "*.stp" format. The stored setups can be recalled.

After selecting this type:

- Press **File Name** to input the file name to be saved. For filename input, refer to descriptions in "To Create a Folder". Press **Storage** → **More** → **Auto Name** to select "ON", then the filename is generated automatically. If the filename that you input is the same as the existing filename, a prompt message will be displayed, reminding you that the existing file will be overwritten and asking you whether to continue or not.
- Press **NewFolder** to create a new storage folder. For details, refer to descriptions in "To Create a Folder".
- Press **Window** to display or hide the disk management interface.
- Press **Save** to save the set setup file.

Note from the manual (p.18-3): the per-channel **storage channel** can only be set when **Data Source = Memory** and the format is **.bin** or **.csv**. The **.wfm** format always stores the full deep-memory record.

1. Press **Save/Recall (or Storage) → Waveform**.
2. **Source**: select the specific analog channel(s) — not “Screen”.
3. **Format — preferred order**: - **.wfm (RIGOL binary)** — exact, smallest, keeps all settings/metadata. Best for archiving and re-loading on the scope. - **.bin** — compact binary, easy to parse in Python/MATLAB (int8/int16 codes + header scale/offset). Good interchange format. - **.csv** — only when a tool truly needs text; expect huge files and 5-sig-fig time rounding. If used, prefer **Save Memory**, never **Save Screen**.
4. **Data source = Memory (deep memory), not Screen**, so you get every sample, not the decimated display buffer.
5. Save to a fast USB-3 stick or pull data over LAN with SCPI (:WAV:SOURce, :WAV:MODE RAW, :WAV:FORMat BYTE, :WAV:DATA?) for automated, repeatable captures.
6. **Record the scaling** (YINcrement, YORigin, YREFerence, XINcrement, XORigin) — these convert raw ADC codes back to volts/seconds. The **.wfm/.bin** headers contain them; CSV bakes them in but loses precision.

7.6 Part E — Hardware/setup hygiene (protect the bits you have)

- **Use the lowest usable attenuation** (1× over 10× when voltages permit): a 10× probe also multiplies noise referred to the input.
- **Short, shielded leads; solid ground**; keep the signal lead away from the drive/VFD power cabling to avoid PWM pickup.
- **Bandwidth-limit on purpose**: if you only care to ~20 kHz, enabling the channel **20 MHz BW limit** (CH → BW Limit) and using High-Res removes out-of-band noise that would otherwise eat codes.
- **Let the scope warm up** and run **Self-Cal** (Utility → Self-Cal) for gain accuracy.
- **Stationarity**: for averaging/long FFTs the machine speed/load must be steady over the record; otherwise sidebands smear. Capture transients as *separate, dedicated* single-shot records in High-Res.

7.7 Quick checklist (tape to the scope)

```
[ ] Probe attenuation set correctly (CH → Probe) [ ] DC coupling [ ] Signal fills
80-90% of screen, NO clipping (incl. transients) [ ] Acquire → Mode → High Res
(Average only if steady & periodic) [ ] Mem Depth fixed; record length ≥ 10-20 s [ ]
Sample rate ~20-50 kSa/s for MCSA (don't waste memory on excess rate) [ ] 20 MHz BW
limit ON if only ≤20 kHz needed [ ] Save → Waveform → Source = channel → Format .wfm
or .bin (Memory, not Screen) [ ] Note Y/X scale & offset (or rely on binary header) [ ]
Bearings/looseness: also grab vibration + acoustic channels
```

7.8 Expected improvement

	Before (our files)	After this procedure
ADC codes used	~82-97 / 256	~200+/256 (fills screen)
Effective bits	≈ 6.7	≈ 10-12 (fill + High-Res)
Aliasing risk	present (Screen)	removed (High-Res band-limit)

File size (25 Mpts)	381 MB CSV	~25–50 MB binary (.bin/.wfm)
Time/scale fidelity	5 sig figs (CSV)	exact (binary header)

7.9 Licensing — is any of this locked? (No)

Every setting in this guide is standard on the MSO5074 and unlocked by default — no license file is required. The optional licenses on the MSO5000 series cover features we do **not** use for motor-current analysis.

Feature used in this guide	Status on MSO5074
Acquisition modes (Normal / Average / High Res / Peak Detect)	■ Standard, free
Deep memory up to 100 Mpts	■ Standard (only the 200 Mpts upgrade is optional)
Waveform export to .wfm / .bin / .csv (deep memory)	■ Standard
SCPI remote + LAN/LXI + built-in web control	■ Standard
Channel 20 MHz bandwidth limit, vertical scale/offset, DC/AC coupling	■ Standard
FFT / Enhanced FFT math, measurements, Self-Cal	■ Standard

Paid option licenses (NOT needed for this work): - Bandwidth upgrade above 70 MHz — irrelevant for 50 Hz signals. - 200 Mpts memory upgrade — 100 Mpts standard is already more than enough. - Built-in arbitrary waveform generator (AWG). - Serial decode/trigger (CAN / LIN / RS232 / I²C / SPI / FlexRay). - Power-analysis application package.

Requires extra hardware, not a license: the 16 digital (logic-analyzer) channels need the RPL logic probe. We use the analog channels, so this does not apply.

Note: menu names/availability depend on **firmware version**, not on any license. If High Res or a deep memory depth seems missing, update to the latest MSO5000 firmware from Rigol — the upgrade is free.

8. Binary exports (.bin) — separate acquisitions

These files were exported in Rigol's **binary waveform** format. They are **not** the same signal as the two CSV files — they are later, independent captures (different amplitude, rate and date), included to validate the binary path and to compare formats.

8.1 What's inside

Property	Value
Format	Binary waveform, RG01 magic, 1 waveform / 1 buffer

Sample encoding	float32 volts (4 bytes/point, exact — not 8-bit text)
Instrument	MSO5074 , serial MS5A262701593
Captured	2026-06-12 14:18:04 (embedded in header)
Samples	1,000,000
Sample interval	50 μ s $\rightarrow$ 20 kSa/s (Nyquist 10 kHz)
Duration	50 s (0 $\rightarrow$ 50 s)
File size	3.81 MB

8.2 Signal statistics (CH1)

Metric	Value
Min	-141.8 mV
Max	5.105 V
Peak-to-peak	5.246 V
Mean (DC offset)	444.9 mV
RMS	1.004 V
Std deviation	899.6 mV
Distinct levels	75
Quantization LSB	70.9 mV (~6.2 effective bits over p-p)
Dominant freq	0.78 Hz (period 1.28 s)

The larger LSB (70.9 mV vs 8.78 mV in the CSV captures) reflects the **coarser vertical scale** used for this ~5 V signal — still the same underlying 8-bit ADC, only ~75 of 256 codes used.

8.3 Format comparison — is binary just "compressed"?

No. For a given acquisition the **samples are identical**, but the binary format is qualitatively better than CSV, not merely smaller:

Aspect	Binary .bin	CSV
Size (1 Mpts)	3.81 MB	~24 MB (Screen-style text)
Numeric precision	exact float32	rounded text (time to ~5 sig figs)
Metadata	model, serial, date/time , units, xInc/xOrig	none / one header line
Load speed	fast (binary slurp)	slow (parse millions of rows)

8.4 Notes on sample rate

- At **20 kSa/s** this binary capture is the **lowest-rate** of the three files (vs 33.3 kSa/s Screen and 500 kSa/s Memory) — but the rate is a function of the **memory-depth/timebase setting at capture**, not the binary format.
- **20 kSa/s is still adequate for 50 Hz MCSA**: Nyquist 10 kHz covers the fundamental, harmonics and the principal fault sidebands. For high-order slot harmonics, raise Acquire → Mem Depth (see §7.4) on the next capture.
- The 8-bit / ~6.2-effective-bit limitation discussed in §6.4 applies here too; the §7 procedure (fill the screen + High-Res) is the way to improve it.

8.5 Additional binary captures (**1v0.bin**, **500.bin**)

Two more binary files, captured ~2.5 minutes apart on the same channel/setup. The signal is nearly identical (~300 Hz, ~2.7 V p-p, ~0.18 V DC); the meaningful difference is **vertical resolution**.

Property	1v0.bin	500.bin
File size	1.91 MB	1.91 MB
Instrument	MSO5074 (S/N MS5A262701593)	same
Captured	2026-06-12 19:47:02	2026-06-12 19:49:32
Samples	500,000	500,000
Sample interval	60 μ s	60 μ s
Sample rate	16.67 kSa/s (Nyquist 8.3 kHz)	16.67 kSa/s
Time window	10 → 40 s (30 s)	10 → 40 s (30 s)
Min / Max	−44.4 mV / 2.706 V	−22.2 mV / 2.748 V
Peak-to-peak	2.75 V	2.77 V
Mean (DC)	175.1 mV	180.2 mV
RMS	362 mV	369.4 mV
Dominant freq	300 Hz	300.2 Hz
Distinct levels	63	126
Quantization LSB	44.4 mV (~6.0 bits)	22.2 mV (~7.0 bits)

Key point — 500.bin is the higher-quality capture. It resolves **twice** the number of amplitude levels (126 vs 63) at **half** the LSB (22.2 vs 44.4 mV), i.e. ~7.0 vs ~6.0 effective bits. This is exactly the gain expected from a **finer vertical scale or High-Res mode** (§7.2–7.3): same signal, same rate, but more of the 256 ADC codes are used. Prefer the 500.bin-style setup for analysis.

Both run at 16.67 kSa/s (500k points ÷ 30 s), which is adequate for the 300 Hz signal and its low-order harmonics; raise memory depth for higher-frequency content.

(BG): 1v0.bin 500.bin (~300 Hz, ~2.7 V)
 12, ~2.5, 30 (10–40 s)
 500 000, 16.67 kSa/s. Sample rate- (Mem Depth)
 (500 000 ÷ 30 s).
 — 500.bin (126 63),
 80–90 %
 Acquire → High Res; sample rate Acquire
 → Mem Depth.

9. Multi-sensor study guidance (current + vibration + sound)

Three independent sensors are available, which is a major advantage: each fault leaves a signature in more than one domain, so **crossconfirmation** turns a "maybe" into a finding and rejects measurement artifacts.

9.1 What each sensor sees

Sensor	File type	What it is	Strength	Limitation
Current (oscilloscope)	.csv / .bin	Armature-current waveform, kSa/s, 8-bit	Sees electrical + load/torque + converter (300 Hz) signatures directly	8-bit vertical (see §6.4); single channel
Vibration (vibrometer)	.XLS	Overall RMS spot values — Velocity (mm/s), Acceleration (m/s ²), Displacement (mm), time-stamped	ISO-10816-comparable trend numbers; direct mechanical evidence	Not a waveform — no spectrum, so it cannot localise a fault frequency on its own
Sound (phone)	.wav	Stereo, 44.1 kHz, 16-bit, ~31 s waveform	Full spectrum to 22 kHz, 16-bit; catches bearing/brush squeal, looseness rattle	Room acoustics, reflections, background noise; not calibrated (relative only)

The .XLS vibration files are **point readings** (one Velocity / Acceleration / Displacement value per timestamp), not time series. They are excellent for **trending vs load** but you **cannot FFT them**. To get a vibration *spectrum* you need the vibrometer's raw time-waveform/FFT export (if it has one) or an accelerometer logged on a spare scope/DAQ channel.

9.2 The golden rule — measure all three at the same operating point

The dataset is already organised by **load (%)**: current, \NN.XLS and \NN.wav all exist for 5, 10, 20 ... 100 %. Keep this discipline:

- **One run = one load level**, with **all three** sensors capturing the **same steady state**. Let the speed settle before recording.

- **Log the operating point** every time: load %, **actual RPM** (tacho), drive firing angle/armature voltage, and bearing/ambient temperature.
- Use a **common, sortable naming scheme** so files line up across folders, e.g. 2026-06-12_load020_run1_current.csv, ..._vib.XLS, ..._audio.wav.
- **Time-align** the records (a shared start marker — e.g. a brief load step or a hand clap visible/audible in all three — makes later correlation trivial).

9.3 Which fault shows where (crossconfirmation map)

Phenomenon	Current	Vibration	Sound	Confirm by
Converter ripple (300 Hz)	strong 300 Hz line, speed-independent	usually absent mechanically	faint 300 Hz hum if at all	present in current at all speeds → electrical, not a fault
Commutation / brush wear	ripple at $S \times \text{RPM}/60$ (~80–120 Hz @200 RPM), broadband arcing noise	rises with wear	brush squeal/chatter , broadband hiss	current band tracks speed + audible squeal
Bearing defects	weak sidebands around line/commutation	clear at BPFO/BPFI/BSF/FTF freqs	high-freq tonal whine/grinding	vibration spectrum is the primary; sound confirms
Imbalance	1× RPM modulation (small)	large 1× RPM velocity peak	low rumble at 1×	vibration 1× dominant, drops if balanced
Misalignment	1×/2× modulation	2× RPM prominent, axial component	—	2× in vibration + axial reading
Mechanical looseness	erratic load current	raised noise floor, many RPM harmonics	rattling	vibration harmonics + audible rattle
Eccentricity / bent shaft	1×/2× modulation	1×/2× with phase shift	—	vibration phase across the machine

Rule of thumb: **electrical-only** signatures appear in current but not in vibration/sound; **mechanical** signatures appear in vibration (and often sound) and only weakly modulate the current.

9.4 Vibration (vibrometer) best practice

- **Mounting matters most.** Rank by usable bandwidth: **stud-mounted ■ glue/ adhesive > magnet ■ hand-held probe**. A hand-held probe is only valid below ~1 kHz — fine for imbalance/looseness, useless for bearings. Use the same mount and the same spot every time.
- **Measure in 3 directions** at each bearing: **horizontal, vertical, axial**. Misalignment shows axially; imbalance shows radially.
- **Use the right quantity: velocity (mm/s RMS)** for general severity (ISO 10816 / ISO 20816 trend), **acceleration (m/s²)** for high-frequency bearing/gear faults, **displacement** for low-speed/shaft issues.
- **Trend, don't spot-judge.** A single number means little; the **change vs the same load over time** is the diagnostic. Plot Velocity-RMS vs load % and watch the curve move between test campaigns.

- If the vibrometer can export a **time waveform or FFT**, use that for bearing-frequency analysis — the overall RMS in the .XLS files cannot localise a defect frequency.

9.5 Sound (phone) best practice

- **Treat it as qualitative/relative.** A phone mic is not calibrated, so use it for **spectral shape and change**, not absolute dB. Keep mic distance and angle fixed between runs (mark a spot on a tripod/stand).
- **Record format:** prefer **WAV/PCM (lossless)**, not compressed MP3/AAC — lossy codecs smear transients and add artifacts. 44.1 kHz / 16-bit is plenty (covers to 22 kHz). Disable any "noise suppression"/"voice enhancement" processing in the recorder app — it destroys machine-noise spectra.
- **Control the environment:** quiet room, no reflective clutter, motor not sitting on a resonant bench; capture a **background-noise reference** (motor off) to subtract/spot ambient lines.
- **Mind aliasing-free range:** 44.1 kHz → 22 kHz usable; good for brush squeal, bearing whine and looseness rattle, which is exactly the audible-fault band.
- **Analyse with a spectrogram** (time × frequency), not just an FFT — bearing and brush faults are often **intermittent**, and a spectrogram reveals bursts that a single averaged FFT hides.

9.6 Test matrix to run

For a defensible study, capture this grid **per machine condition** (healthy baseline first, then each seeded/observed fault):

1. **Load sweep:** 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 % — all three sensors at each step, steady state. (You already have this layout.)
2. **Speed/RPM logged** at every step (don't assume linear 20 %→400 RPM; the thyristor drive's speed depends on firing angle and load — measure it).
3. **Transients (current + sound, scope in single-shot):** start-up, sudden load step, **reversal** (unipolar on this rig — current stays ≥ 0 , so the normal 0-at-bottom scale is fine), and stop/coast-down.
4. **Repeat x3** at each key point to separate real effects from run-to-run scatter; report mean $\pm$ spread.
5. **Baseline vs fault:** the whole value is in the **difference** — always have a healthy reference recorded under identical settings to compare against.

0000000000 (BG): 0000 000 00000000 — 0000 (waveform), 00000000 (overall RMS
00000000 0 .XLS, 000 0000000) 0 0000 (.wav, 44.1 kHz, 16-bit). 000000 0 00000 0
0000 0 0000 00000000 00000 (00000 000000000000, 000000000 00000) 0
00000000 00000000 0000000 00000 000. 000000000 00000000000: 300 Hz
0 0000000000 (0000 00000 0000, 00 000000 00 000000000) → 00 0 000000;
00000000000 0000000 (0000000000, 000000000000, 000000, 00000000000)
00 000000 000-00000 0000 0000000000 (00000 0 3 000000, velocity mm/s 00 ISO
10816) 0 00 000000 0 00000. 00 00000000 00000000 00 0000000 waveform/FFT 00
000000000 — overall 0000000000 0 .XLS 00 0000000000 000000.
00000000 0 00000000000 (000000000000): 0000 0 WAV, 00000000
0000000000000, 00000 00 00000000 000000000000 0 000000 000000000000.
0000000 00000 00000 baseline 00 0000000000.

10. Native `.wfm` deep-memory exports — and are they better than `.bin`?

Two new files were added: `normal.wfm` and `normal_scaleddown.wfm`. The script now parses the Rigol MSO5000 `.wfm` format (`python analyze_rigol.py normal.wfm normal_scaleddown.wfm`).

10.1 What is inside a `.wfm`

Property	Value
Layout	4200-byte proprietary header + raw 8-bit ADC codes, 1 byte/sample
Samples	25 000 000 (full deep memory)
File size	23.85 MB (vs ~380 MB for the same points as CSV)
Vertical LSB	857.9 μV/code (read from header)
Codes used	84 distinct , range 0–186
Sample rate	not stored in any documented field — assumed 500 kSa/s (25 M $\div$ 50 s); override with <code>WFM_FS</code>

■ The `.wfm` **horizontal scale and absolute vertical offset are not openly documented**, so the script decodes the **vertical LSB and the raw codes exactly** (resolution + clipping are trustworthy) but **assumes** the time base. Pass the real rate with `WFM_FS=<Sa/s>` if it differs.

10.2 The two files are the *same* capture

`normal.wfm` and `normal_scaleddown.wfm` are **practically identical**: out of 25 000 000 samples only **4 differ** (correlation 0.9999989), the vertical LSB is the same (857.9 μ V), and the code range is the same (0–186). The `scaleddown` name does **not** correspond to a finer vertical scale — there is no resolution difference between them. Treat them as duplicate exports of one acquisition.

10.3 The code-0 block is an *intentional* zero-current baseline (not clipping)

20 % of each record (4 999 949 of 25 M samples, a contiguous ~10 s block at the **start**) sits at code 0. This is **deliberate and correct**, not a fault: this is a **unipolar DC-motor current**, so 0 was placed at the **bottom of the screen** (the offset-to-bottom strategy of §7.2a/§7.2b) to use the full ADC range for the positive-only current. The code-0 block is simply the **zero-current period before the motor was energised** — there is no signal there to lose.

The data confirms it is a baseline, not clipping:

- The code-0 samples form **one contiguous block at the very start** (~0–10 s), not scattered through the running record.
- Once the motor runs, the signal **keeps clear headroom above the bottom rail**: its **1st percentile is code 62 of 255** and there are **no code-0 samples** during operation. Nothing is railed.
- So the deliberate “0 at the bottom” worked exactly as intended: full positive range used, no negative excursions lost (the motor runs unipolar — no reversal/regeneration in this capture).

Reversal is also unipolar on this rig — no bipolar record needed. The controller **keeps the current ≥ 0 even in reverse** (verified: at 1 V/div with **100 % nominal load and reverse** the trace still fits on screen and

never hits either rail). The small negative readings in the data (e.g. `1v.bin` `min` ≈ -0.31 V) are **measurement noise/offset around zero**, not real reverse current, and they are fully captured because the baseline sits just above the bottom rail. So the “0 at the bottom” unipolar scaling is valid for **every** regime here. (Only a drive that produced genuine negative/regenerative current would need a bipolar, zero-centred scale — not this one.)

Analysis note. The long flat baseline injects a large DC step that **dominates the FFT** (the script reports a spurious ~ 0.02 Hz “50 s period” line) and hides the real 300 Hz. This is *not* a data problem — just **analyse only the running window** (skip the first ~ 10 s) and the true spectrum returns. The updated script now recognises this pattern and prints a *zero-current baseline* note instead of a false clipping warning.

10.4 Are `.wfm` better than `.bin`?

It depends on what “better” means — they win on **completeness**, not on resolution:

Aspect	<code>.bin</code> (RigolDS0)	<code>.wfm</code> (normal)	Winner
Encoding	float32 volts, 4 B/sample	raw 8-bit codes, 1 B/sample	<code>.wfm</code> (4× denser)
Points kept	1 M (screen-decimated)	25 M (full deep memory)	<code>.wfm</code>
Bytes per million pts	~4 MB	~1 MB	<code>.wfm</code>
Vertical resolution	8-bit (~75 levels)	8-bit (~84 levels)	tie (both 8-bit)
Self-describing	yes — volts, time, model, serial, date	no — undocumented time base/offset	<code>.bin</code>
Ease of use	load & go (already volts)	needs this custom parser; rate assumed	<code>.bin</code>
This capture's quality	clean	clean (intentional zero baseline, no clipping)	tie

Bottom line. - The `.wfm` **format** is the better *archival* choice when you want the **entire deep-memory record** in the **smallest** file: 25× more samples than the 1 M `.bin`, at $\sim 1/4$ the bytes per sample. - But `.wfm` is **not** higher vertical resolution — it is the **same 8-bit** data, and it is **not self-describing** (the time base and vertical offset are undocumented, so any tool must assume them). - For *this* dataset both captures are clean: the `.wfm` code-0 block is the **intentional zero-current baseline** (§10.3), not clipping. The `.bin` is still easier to use because it is self-describing (volts + time + metadata). - **Recommendation:** use `.wfm` for compact full-depth archiving, and **always record the time/div and sample rate** alongside it, since the file itself does not preserve them in a documented way. For an analysis that must be self-contained and unambiguous, prefer `.bin` (or Save Memory CSV).

```
##### (BG): normal.wfm ■ normal_scaleddown.wfm ■■ #####
(##### 4 ##### 25 #####). ##### .wfm #####
##### — 25 #####. ##### 24 MB (8-##### 1 #####/#####), #####
■ -##### ■ .bin, ■ ■ ##### 8-##### ■ ■
##### (##### ■ #####)
##### → sample rate ■ ##### 500 kSa/s, override ■ WFM_FS). #####
##### 0 (~10 s ■ #####, 20%) ■ ##### — ##### DC #####
```

0 (7.2a/7.2b), 0
; 0
0. 0: 0
(1-0 = 62 255) 0. 0: 0
FFT (0.02 Hz), 0
(0.10 s). 0: 0
→ 0
.

11. Vertical-scale experiment — 1v.bin vs 2v.bin (1 V/div vs 2 V/div)

Two new captures were taken on 2026-06-16 with the *only* deliberate change being the **vertical scale**, plus photos of the scope settings (1v.jpg, 2v.jpg).

11.1 Settings read from the photos

	1v.jpg → 1v.bin	2v.jpg → 2v.bin
CH1 scale	1.00 V/div	2.00 V/div
CH1 offset	−3.00 V	−6.00 V
Time base	25 s window, 1 MSa/s, 20 Mpts	25 s window, 1 MSa/s, 20 Mpts
Trigger	CH1 edge, 728 mV	CH1 edge, 728 mV
Source / format	Memory → .bin	Memory → .bin
Captured	15:48:41	15:50:29

Both the scale **and** the offset were doubled together (1→2 V/div, −3→−6 V), which keeps the trace in the same place on the screen.

11.2 What the data shows

Metric	1v.bin	2v.bin	Relationship
Samples	20 000 000	20 000 000	same
Sample rate	1 MSa/s	1 MSa/s	same
Window	−15 ... +5 s	−15 ... +5 s	same
Min / Max	−0.311 V / 5.057 V	−0.622 V / 10.124 V	exactly ×2
Mean / RMS	0.517 V / 1.117 V	1.035 V / 2.237 V	exactly ×2
Quantization LSB	44.36 mV	88.81 mV	exactly ×2
Distinct levels	122	122	identical

(BG): 1 V/div (100% ΔV), 100% ΔV (100% ΔV), 100% ΔV (≥ 0) 1 V/div 100% ΔV $\rightarrow$ 100% ΔV (100% ΔV 100%+ ΔV) — High-Res (7.3) 1 V/div.

Load	Accel (mean, m/s^2)	Velocity (mean, mm/s)
5 %	1.0	~0
10 %	5.0	~0
20 %	7.7	~0
30 %	12.2	~0.01
40 %	17.0	~0
50 %	17.9	~0
60 %	32.7	0.09
70 %	26.1	0.06

80 %	44.1	0.22
90 %	37.1	0.11
100 %	35.9	0.12

- **Acceleration is the meaningful channel** here: a clean, near-monotonic climb from ~1 to ~44 m/s² as load increases — exactly the load-dependent mechanical excitation you expect. The peak is around **80–90 %**.
- **Velocity reads ≈ 0** at low/medium load and only lifts at high load. These are **overall RMS spot values**, not a spectrum, so they can trend severity but **cannot localise a fault frequency** (no bearing-frequency analysis).
- The vibrometer evidently outputs **acceleration** most reliably; treat the Velocity/Displacement columns as secondary (they look integrated/quantised).

12.2 Sound (■■■■/* .wav) — what is inside

Format: **stereo, 44.1 kHz, 16-bit, ~31 s** per file, one per load (1, 2, 5...100 %).

■■ **Finding 1 — the two channels are NOT mic vs piezo; they are the same signal.** Across all files the left/right correlation is **0.99999** and $R \approx 1.01 \times L$ (a fixed ~1 % gain difference). So both channels carry the **same recording** — the intended separation of *microphone sound* vs *direct piezo analog* did **not** end up on the two channels. If the goal is to capture both independently, route **piezo → Left, mic → Right** through a true stereo/2-channel interface and re-verify that L and R decorrelate.

■■ **Finding 2 — the recorder clips at high load (AGC overload).** Peak hits **full scale (1.000)** in every file, and the sustained clipping fraction grows with load:

Load	clip %	dominant freq
1–5 %	0.00 %	300 Hz (converter ripple, audible)
10–40 %	~0.01–0.03 %	2.4–2.7 kHz
50–70 %	0.03–0.13 %	5.3–6.2 kHz
80 %	0.41–0.59 %	5.3 kHz
100 %	0.33–0.50 %	6.6 kHz

- At **light load the dominant tone is 300 Hz** — the same mains-locked converter ripple seen in the current (§7.2b), here radiated as audible hum.
- At **higher load the dominant shifts into the kHz range** (mechanical/acoustic noise, brush/structural), and RMS rises with load — but the **clipping at 80–100 % corrupts the spectrum** there (clipping manufactures false harmonics). **The phone gain was too high.**
- 44.1 kHz/16-bit bandwidth itself is fine; the problem is **level/AGC**, not the format.

12.3 The 1 m phone-recording idea — assessment

Adding a **phone at 1 m** is worth doing as a *cheap, non-contact, repeatable* acoustic channel — **but only with fixed settings**, and **as a complement**, never a replacement for the contact piezo/vibrometer:

Good for: trending overall loudness vs load, catching brush squeal/bearing whine, broadband spectral *shape changes* between healthy and faulty, quick go/no-go screening.

Limitations to control: - **Not calibrated** → relative only (spectral shape and change, not absolute dB). - **Far-field at 1 m** picks up **room reflections and background noise**; SNR is much worse than the contact piezo. Use a quiet room, fixed geometry. - **Phone AGC/“noise suppression” must be OFF** — exactly what clipped the existing files. Lock the gain so the **loudest case (100 %) peaks at ≈ -6 dBFS (≈ 0.5 full-scale), never 0 dBFS**. - **Lossless WAV/PCM only** (no MP3/AAC).

Verdict: keep the **contact piezo/vibrometer** as the **primary** mechanical sensor (best SNR, structure-borne), and use the **1 m phone** as a **secondary far-field acoustic reference** for cross-confirmation (§9.3). The two together separate **structure-borne** (piezo) from **airborne** (phone) signatures.

12.4 Recommended nominal loads to investigate

From the data, vibration/acoustic energy is **strongly load-dependent**, so sweep the full range but emphasise the informative points:

- **Mandatory baseline: 0 % / idle** (motor energised, no load) — the healthy reference every comparison needs.
- **Full sweep:** 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 % (you already have this — keep it; it is what revealed the monotonic accel trend).
- **Emphasis around 80–90 %**, where acceleration and acoustic RMS **peak** — the most fault-sensitive operating point; take **≥ 3 repeats** here.
- **100 % nominal** — worst-case electrical/thermal; repeat ×3.
- **Transients (separate captures):** start-up/inrush, a **load step** (e.g. 50 % → 90 %), **reversal**, and coast-down — these stress bearings/brushes most and are where incipient faults first show.
- **Light load 1–5 %** is useful too: there the **300 Hz converter ripple dominates**, giving a clean electrical reference uncontaminated by mechanical noise.

12.5 Example experimental setup (■■■■■■■■ ■■■■■■■■■■)

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1. **CH1**: **MSO5074**, **1 V/div** (**100 %+**, §11.4), **DC coupling**, **Acquire** → **High Res**, **Memory** → **.bin**, **1 MSa/s**, **20 Mpts**. 2. **(****stud****)**: **3** **(****stud****)**; **Acceleration (m/s^2)** **1 m**, **AGC/**; **WAV/PCM 44.1 kHz**; **100 %** **≈ -6 dBFS**. 3. **2-** **(****L****R****)**. 4. **RPM**

..... (setup.jpg).
4
..... „..... ↔” (fixed everything else).

12.6.1 (4 , 20 s)

..... (§12.5):

#		 →			
1	 (structure-borne)	 (.....), →		20 s	WAV/PCM,, AGC OFF
2	 (air-borne)	,	 1 m,	20 s	WAV, AGC/..... OFF, peak ≈ −6 dBFS
3		 → .XLS (overall RMS)		20 s	Acceleration; ~0.5 s/.....
4		 →	/..... s		D 25.05 (.....), 1.00 V/div, offset 2 V, High Res, → .bin

..... 20 s: $\Delta f = 1/20 = 0.05$ Hz; 200 RPM (§).
.....
(..... #4 «−1.00 V» = 1.00 V/div §11.4; «2V» = 2 V)

..... 1 2:
..... (..... 0.99999) —
.....: →
...../..... → (.....), L≠R
.....

12.6.2 (.....)

..... A — (.....) — baseline „.....” - A1. ,
.....; - A2. ■

§12.5: — 4 s — ~5 20 s (~1–2 s); 6–8 s/1 V/div; unipolar ≥ 0 .

B — (mechanical looseness) - , / (: . / , - B1. (A1). - B2. (A2, 4 s). - : (), structure-borne , (looseness 1×, 2×, 3×... RPM); . (/).

C — (control faults) - 14XXX (). (): - / (,). - / . → - , ; 300 Hz / . - / , . - (). - 4 () + 300 Hz .

12.6.3

A/B/C - () :

A	A1 , A2 /4 s	0 % (idle) + sweep 5–100 %, 80–90 %	×3 80/90/100 %
B	B1 , B2 /4 s	A	×3
C		(. 50/80/100 %)	×3

: A, B C () — (, , ,) .

12.6.4

- : 2026-06-16_{A|B|C}_{sub}_{loadNN}_runN_{vibmic.wav,phone.wav,vib.xls,current.bin} 2026-06-16_B_revers_load090_run2_current.bin.

- **Acceleration-Displacement:** Acceleration is proportional to Displacement / Frequency. Acceleration is proportional to Displacement, and Displacement is proportional to Acceleration.
- **Acceleration Displacement Ratio (ADR) (Displacement Ratio)** is proportional to Acceleration and Displacement.

12.6.5 Acceleration vs Displacement Ratio

Parameter	Acceleration Ratio	Displacement Ratio
Acceleration Ratio (B)	Acceleration + Acceleration (accel) + Displacement 1 m	↑ overall accel; Displacement 1×/2×/3× RPM, Displacement Ratio ↑
Displacement Ratio (C)	Displacement (Displacement) + 300 Hz Displacement	Displacement/Displacement Displacement, ↑ 300 Hz Displacement Displacement Displacement (Displacement)
Displacement (A)	Displacement	Displacement 300 Hz (Displacement.) + Displacement ∝ RPM; Displacement Displacement

- **ADR:** Acceleration Ratio is proportional to Displacement Ratio (Displacement Ratio), FFT; Displacement Ratio is proportional to A↔B↔C Displacement Ratio is proportional to Displacement.
- **Acceleration:** Acceleration vs Displacement is proportional to Displacement Ratio (Displacement Ratio A/B/C). Displacement Ratio/looseness Displacement Ratio waveform/FFT Displacement Ratio, Displacement is proportional to Displacement (overall RMS Displacement Ratio).
- **ADR:** Acceleration Ratio is proportional to Displacement Ratio (A/B/C Displacement Ratio); Displacement Ratio.
- **Displacement Ratio Ratio (\$9.3):** 300 Hz = Displacement Ratio; Displacement Ratio (looseness) → Displacement+Displacement; Displacement Ratio → Displacement+300 Hz.

Displacement Ratio (BG): Displacement Ratio is proportional to Displacement Ratio (Displacement Ratio 20 s) Displacement Ratio Displacement Ratio: A) Displacement Ratio (Displacement Ratio + Displacement Ratio/4 s), B) Displacement Ratio Displacement Ratio (Displacement Ratio + Displacement Ratio), C) Displacement Ratio Displacement Ratio Displacement Ratio (Displacement Ratio Displacement Ratio 14XXX Displacement Ratio). Displacement Ratio B Displacement Ratio Displacement Ratio/ Displacement Ratio/Displacement Ratio (↑ Displacement Ratio, Displacement Ratio Displacement Ratio); Displacement Ratio C Displacement Ratio is proportional to Displacement Ratio (Displacement Ratio Displacement Ratio, ↑ 300 Hz). Displacement Ratio Displacement Ratio Displacement Ratio A/B/C, Displacement Ratio Displacement Ratio Displacement Ratio — Displacement Ratio Displacement Ratio Displacement Ratio. Displacement Ratio 1 V/div, AGC OFF, Displacement Ratio Displacement Ratio Displacement Ratio Displacement Ratio.

13. Acceleration vs Displacement Ratio (Displacement Ratio + Displacement Ratio Displacement Ratio)

Displacement Ratio Displacement Ratio Displacement Ratio Displacement Ratio. (Displacement Ratio — Displacement Ratio Displacement Ratio Displacement Ratio.)

13.1 Acceleration vs Displacement Ratio

- **C1. Cross-channel** [redacted] [redacted]: [redacted] $\leftrightarrow$ [redacted] ([redacted] structure-borne [redacted] [redacted] air-borne), [redacted] $\leftrightarrow$ [redacted] ([redacted] [redacted]), [redacted] $\leftrightarrow$ [redacted]@300 Hz ([redacted], [redacted] 300 Hz [redacted] [redacted], [redacted]).
- **C2. [redacted]-[redacted]** [redacted] ([redacted] [redacted] Pearson): [redacted] [redacted]
\$\\gamma^2(f)\$ ([redacted] [redacted] [redacted] [redacted] [redacted]), envelope
[redacted] (Hilbert $\rightarrow$ [redacted]; [redacted] [redacted] [redacted]),
cross-correlation/lag ([redacted] $\leftrightarrow$ [redacted] ~3 ms [redacted] 1 m).
- **C3. [redacted] vs [redacted]:** [redacted] [redacted] [redacted] 5 $\rightarrow$ 100 % ([redacted]
[redacted]-[redacted] SNR [redacted] [redacted]).
- **C4. [redacted] [redacted] feature-[redacted]** [redacted] A/B/C: RMS, kurtosis, crest factor, 1 $\times$ /2 $\times$ /3 $\times$ RPM, 300 Hz,
[redacted] [redacted] $\rightarrow$ [redacted] [redacted] [redacted]-[redacted] [redacted]
[redacted] (Fisher-[redacted] score, [redacted] ML).
- **C5. Heatmap** [redacted] N $\times$ N [redacted] [redacted]/feature-[redacted] [redacted] [redacted]; [redacted] [redacted]
A/B/C [redacted] [redacted] [redacted] [redacted] [redacted].

- R1. Reference vs candidate: [REDACTED] = [REDACTED] ([REDACTED]/[REDACTED], [REDACTED], [REDACTED]); [REDACTED] = [REDACTED] (1 m [REDACTED]; [REDACTED] [REDACTED]).
- R2. [REDACTED] [REDACTED]: [REDACTED] [REDACTED] looseness (B) → [REDACTED] 1x/2x/3x RPM; [REDACTED] [REDACTED] control fault (C) → ↑ 300 Hz/[REDACTED] [REDACTED]; [REDACTED] [REDACTED] [REDACTED]; [REDACTED] [REDACTED] [REDACTED].
- R3. [REDACTED] [REDACTED] [REDACTED]: [REDACTED] (ROC/[REDACTED] — [REDACTED] [REDACTED] [REDACTED] [REDACTED] B [REDACTED] C [REDACTED] A), Spearman [REDACTED] [REDACTED] ↔ [REDACTED] [REDACTED] [REDACTED], [REDACTED] [REDACTED] ↔ [REDACTED] [REDACTED] [REDACTED] ([REDACTED] [REDACTED] [REDACTED]/[REDACTED]), [REDACTED] [REDACTED] SNR/[REDACTED] [REDACTED] ([REDACTED] [REDACTED]).
- R4. [REDACTED] [REDACTED]: [REDACTED] [REDACTED] [REDACTED] [REDACTED]/[REDACTED] [REDACTED] [REDACTED] [REDACTED] 300 Hz [REDACTED]; [REDACTED] [REDACTED] [REDACTED]/[REDACTED] [REDACTED] [REDACTED], [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]; [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]/AGC → [REDACTED] [REDACTED]: [REDACTED] [REDACTED] [REDACTED], [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]. [REDACTED]: [REDACTED] [REDACTED] [REDACTED] [REDACTED] („[REDACTED] X %, [REDACTED] Y").
- R5. [REDACTED]: [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] ([REDACTED] [REDACTED]) [REDACTED] [REDACTED] — [REDACTED] [REDACTED] [REDACTED] [REDACTED]: [REDACTED] [REDACTED] (1 m) [REDACTED]-[REDACTED].